

# Preston H. Carpenter

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## EXPERIENCE

**Stripe** | Seattle, WA

June 2025 - Present

*Senior Software Engineer*

- Created an agentic workflow that analyzed code to classify IAM roles, applied classification-specific prompts to determine required permissions, and autonomously created pull requests with the resulting code changes.
- Designed a deterministic validation gate for AI-generated changes, replacing unreliable LLM self-review with programmatic checks that hard-stopped invalid changes.
- Added orphaned-resource cleanup to a reconciliation system designed to replace Terraform IaC across the company, eliminating hundreds of hours of toil cleaning up resources left behind by infrastructure drift.
- Unblocked a project to split out deployment targets, which required deploying multiple 15 gigabyte container images, by designing and implementing a dynamic image pulling daemon using containerd and Go to stream images onto shared hosts.
- Migrating a legacy Airflow execution layer from dedicated hosts to dynamic pod workers in Kubernetes with a projected infrastructure savings of millions of dollars.

**Amazon Web Services (AWS)** | Seattle, WA

April 2020 - June 2025

*Software Engineer II*

- Led the redesign of Amazon Linux's global release system, migrating a 4-day, console-driven process to AWS Step functions and CDK, reducing release time by 94%, to 6 hours.
- Built a Python CLI that codified the Amazon Linux release process, replacing an ad-hoc console workflow and tribal knowledge with a repeatable, self-service engineering tool.
- Owned production releases across all AWS regions and partitions, coordinating releases of AL1, AL2, AL2023, and internal variants under strict deadlines.
- Maintained the security of globally distributed Amazon Linux releases, patching critical CVEs and participating in the chain of trust for embargoed security releases including the Linux kernel, `sudo`, and `log4j`.
- Automated operational workflows across AWS partitions using Rust, Python, and Bash.
- Mentored engineers across the team, including managing an intern who converted to full-time and mentoring a junior engineer into a more technical role.

**Starry** | Boston, MA

February 2019 - April 2020

*System Software Engineer*

- Implemented backbone fiber network fail-over with Python and networkd.
- Designed and implemented SSH-based remote device support system for technicians, using Rust and gRPC.
- Wrote Rust and C++ firmware for embedded devices that power Starry internet.

**Google** | *Software Developer Intern* | Montreal, QC

August 2018 - November 2018

**Microsoft** | *Software Engineer Intern* | Redmond, WA

May 2018 – July 2018

**Intuit** | *Software Engineer Co-op* | San Diego, CA

July 2017 – December 2017

**Kinto Care** | *Full Stack Engineer Co-op* | Boston, MA

July 2016 – January 2017

**Beechwood Software** | *Software Engineer Intern* | Boston, MA

January 2015 – July 2016

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## SIDE PROJECTS

**Fuel Crisis (2026 GMTK Game Jam Entry)**

July 2026

- Designed, programmed, and animated a complete 3D puzzle game in Godot and Blender for the GMTK Game Jam; ranked in the top 20% of 10,000+ entries

**Way Cooler (Wayland Window Manager)**

July 2016 – January 2020

- Independently created tiling Wayland window manager with 6,000+ downloads and 2,000+ stars on GitHub.
- Designed a programmable interface using Lua that extends a basic compositor core in Rust.

## COMPUTER KNOWLEDGE

**Languages:** Python, Rust, Go, Java, JavaScript, TypeScript, C, C++, Bash, Lua

**Cloud & Infra:** AWS, CDK, Kubernetes, Docker, systemd

**Distributed Systems:** Step Functions, gRPC, Airflow

**Databases & Web:** PostgreSQL, Flask, NGINX

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## EDUCATION

**Northeastern University** | Boston, MA

September 2014 – May 2019

*Bachelor of Science in Computer Science*